INCH-POUND

MIL-E-63399B(AR)
AMENDMENT 7
27 JUNE 1990
SUPERSEDING
AMENDMENT 6
7 June 1988

MILITARY SPECIFICATION

ELECTRONICS-MCD ASSEMBLY
FOR
MINE, ANTITANK: HE, M75

This Amendment forms a part of Military Specification MIL-E-63399B(AR), dated 16 November 1981, and is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1.1, SPECIFICATIONS:

MILITARY: Add the following:

"MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging
MIL-P-14232 - Parts Equipment and Tools For Army Materiel, Packaging of
MIL-S-19500 - Semiconductor Devices, General Specification For"

PAGE 4

PRODUCT AND PACKAGING DRAWINGS: Delete:

"7258943 - Shipping and Storage Container, M548."

PAGE 7

3.5.3.8a: Delete in its entirety.

3.5.3.8b, line 2: Delete "2.5 Volts" and substitute "V_B-.3 Volts".

AMSC N/A

FSC 1345

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PAGE 9

3.5.7.1 Solderability: Delete in its entirety and substitute:

"3.5.7.1 Solderability. Electronic and electric assembly parts that are to be soldered shall comply with the solderability requirements of MIL-S-45743 and MIL-S-46844 except integrated circuits shall be tested for solderability using the methods and acceptance criteria of MIL-STD-883 and diode and transistors shall be tested for solderability using the methods and acceptance criteria of MIL-STD-750 provided the leads are not gold plated. When gold plated leads are used, the provisions of MIL-S-45743 apply. For assembly parts without gold plated leads, Method 208 of MIL-STD-202 may be used as an alternate to that specified in MIL-STD-750 or MIL-STD-883."

3.5.7.2, line 9: Delete "ARRADCOM, DRDAR-QAT-M" and substitute "ARDEC, AMSMC-QAT-M (D)".

3.7 First Article Inspection: Add at the end of the paragraph "(see 4.3 and 6.2)."

3.8 Surveillance units: Change "Electronic - MCD" to "Electronics - MCD".

PAGE 10

4.3 First Article Inspection: Add after title "(see also 6.2)".

4.3.1:

Change "Lid, Electric Assembly" to "Lid, Electronics Assembly"

Change "Housing, Electic Assembly" to "Housing, Electronics Assembly".

Change "Pinted Wiring Board" to "Board, Printed Wiring".

Drawings: Delete "9345328" and substitute "9292256".

PAGE 12

TABLE I, line 1: Delete "Dwg. 9345328" and substitute " Dwg. 9292256".

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PAGE 13

Table I:

Magnetometer Assembly: Delete "9292146" and substitute "9287146".

Lines 3, 4, 5, 6: Following test(s) add "2/".

Shorting Bar and Connector Assembly: Delete "4.4.2.46 and substitute "4.4.2.45".

Notes: Add "2/ These tests may be omitted at this sub-assembly if performed on Electronic Assembly, DWG. 9292223."

PAGE 15

4.4.2.1: Delete page 15 (Dwg. 9345328) in its entirety and replace with the included new page 15 (Dwg. 9292256).

PAGE 16

4.4.2.2:

Major 101: Delete "Any component . . . assembled, 1/" and substitute "Any component damaged, missing or improperly oriented or assembled".

Majors 102, 103 and 104: Delete in their entirety.

Majors 105, 108, 109, 110, Inspection Method Column: Add 1/.

Minor 201: Delete in its entirety.

NOTES: Change note 1/ to read: "Inspection shall be performed prior to assembling subassembly of shorting bars and power supply to Printed Board Assembly."

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Board, Printed Wiring		9292256
			NEXT HIGHER ASSEMBLY 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	O.D. of board		0.40%	3.2	Gage
102	Center of hole diameter		0.40%	3.2	Gage
110	Any discontinuous front circuit		100%	3.2	Tester
111	Any discontinuous rear circuit		100%	3.2	Tester
112	Solderability	10	10-0-1	3.5.7	4.5.18, 4.5.20
113	Plated through holes integrity			3.2	Visual
114	Visual and dimensional examination			3.2	MIL-P-55110
115	Diameter of component/terminal holes	1/		3.2	Gage
Minor					
201	Contamination on board, 2 sides		0.65%	3.2	Visual, 4.5.18
202	Poor workmanship		0.65%	3.9	Visual

NOTES:
1/ Printed Wiring Board sample plan per MIL-STD-105, AQL 0.40%, Inspection Level S4. From each board, randomly select and inspect 32 hole diameters. A board shall be classed defective if any hole is defective.

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* 4.4.2.3:

Major 101: Delete in its entirety.

Critical 1: Delete in its entirety and substitute: "None defined".

Major 109: Delete "1/".

Note 1: Delete in its entirety.

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PAGE 19

* 4.4.2.4:

Majors 108 and 109, Requirement Column: Delete "3.5.3.11a and 3.5.3.11b" and substitute "3.5.3.11".

Minor 201: Delete in its entirety and substitute: None defined.

PAGE 21

* 4.4.2.6:

Critical 2: Add Incorrect system lens, GATOR vs GEMSS (9 to 12). 100% 3.2 Tester

Major 102: Delete in its entirety.

Major 109: Change "(9-12)" to "(9 to 12)". Change "Visual" to "Ohmmeter".

PAGE 22

4.4.2.7: Delete Minors 203 and 204.

PAGE 23

* 4.4.2.8: Delete in its entirety.

PAGE 24

* 4.4.2.9:

Majors: Delete 104, 105, 106, 107 and 108 in their entirety.

Major 109: Delete in its entirety.

NOTES: Delete in its entirety.

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PAGE 28

4.4.2.13, Drawing Number: Delete "9292151 P/N 9292151-1/-2" and substitute "9287177/9292151".

PAGE 30

* 4.4.2.15: Delete in its entirety and substitute new page 30a including 4.4.2.15.

PAGE 35

4.4.2.20:

Major 104: Delete "3.2" and substitute "3.5.7.1".

Major 105: Delete "6/".

Major 106: Delete "all entries" and substitute "Internal visual - 6/ - LTPD 10% - 3.2 - MIL-STD-750, Method 2072, Para. 3.2 only".

Notes, 6/: Delete "note 6/" and substitute "6/ Sample per MIL-S-19500, APPENDIX C."

PAGE 36

4.4.2.21:

Major 104: Delete "3.2" and substitute "3.5.7.1".

Major 105: Delete "6/".

Major 106: Delete "all entries" and substitute "Internal visual - 6/ - LTPD 10% - 3.2 - MIL-STD-750, Method 2072, Para. 3.2 only".

Notes, 6/: Delete "note 6/" and substitute "6/ Sample per MIL-S-19500, APPENDIX C."

PAGE 39

4.4.2.24, Drawing Number: Delete "9292138" and substitute "92187175".

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Fuse, Electric	1 OF 1	9328688
			NEXT HIGHER ASSEMBLY: 9292223

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Fuse resistance		100%	3.2	Milliohmmeter, 2/
102	Case to lead insulation resistance		100%	3.2	Megohmmeter
103	No fire current for 10 second period, min.	1/	0.40%	3.2	Tester
104	Bridge wire material		100%	3.2	Certification
105	Fuse dimensioning		0.40%	3.2	Gages
106	All fire energy (capacitor discharge) test		0.40%	3.2	Tester
107	Solderability	10	10-0-1	3.5.7.1	MIL-S-45743
108	Evidence of etching compound or contaminants	3/	--	3.2	Chemical analysis
Minor					
201	Poor workmanship		0.65%	3.9	Visual

NOTES:
1/ Use these units for Major 106: All fire energy test.
2/ Limit test current to 1 ma.
3/ One (1) foot from each spool of bridge wire material

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PAGE 41

4.4.2.26: Delete in its entirety.

PAGE 42

4.4.2.27, Major 103: Delete "3.2" and substitute "3.5.7.1".

PAGE 43

4.4.2.28, Major 105: Delete "3.2" and substitute "3.5.7.1".

PAGE 44

4.4.2.29, Major 102: Delete ".5" and substitute "5".

PAGE 45

4.4.2.30:

Title: Delete and substitute "Integrated Circuit, Single CMOS, Target Sensor".

Drawing Number: Delete "9287124" and substitute "9287173".

Major 102: Delete ".5" and substitute "5".

Delete "Visual" and substitute "2/".

PAGE 46

4.4.2.31, Major 102: Delete ".5" and substitute "5".

PAGE 47

4.4.2.32, Major 102: Delete ".5" and substitute "5".

PAGE 54

Major 101: Delete "Diameter of EBP lead, 2 places" and substitute "Diameter of hole for EBP lead, 2 places".

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PAGE 55

4.4.4.40, Drawing Number: Delete "9292264" and substitute "9292624".

PAGE 56

* 4.4.2.41: Delete 9332372 and substitute 9332372/9378309.

PAGE 57

* 4.4.2.42:

Major 101: Delete "Profile of MCD assembly nest" and substitute: "Contour of MCD assembly nest".

Minor 204: Delete in its entirety.

Minor 208: Delete "Thickness at ... nest" and substitute: "Depth of MCD assembly cavity".

Delete 9292243 and subsitute 9292243/9378308.

PAGE 62

4.4.2.47, Power Supply Assembly, P/N 9292241: Add the following Major Categories:

"103, Battery Leaks, 3/ 3.2 4.5.29.1".

NOTES: Add "3/ Sample size per para. 4.5.29.1".

PAGE 63

4.4.2.48:

Major 101: Delete.

Major 102: Delete "Conductive . . . liners" and substitute "Any defect of barrier bag".

Major 103: Delete "5.1" and substitute "5.1, 5.2".

Major 104: Delete.

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PAGE 65

4.4.5:

Line 30: Delete "ARRADCOM" and substitute "ARDEC".

Line 31: Delete "DRDAR-QAT-I" and substitute "AMSMC-QAT-I(D)".

Note 1, line 6: Delete "ARRADCOM" and substitute "ARDEC".

Note 1, line 7: Delete "DRDAR-QAT-M" and substitute "AMSMC-QAT-I(D)".

PAGE 68

4.5.9, line 7: Delete "and spurious outputs".

4.5.12, line 1: Delete "elecronic" and substitute "electronic".

PAGE 69

4.5.20, line 2: Delete "to the Government" and substitute "for Government approval".

PAGE 71

4.5.26 subpara f: Change "Electronic-MCD" to "Electronics-MCD".

* 4.5.28, 4.5.28.1, 4.5.28.2, 4.5.28.3: Delete in their entirety.

PAGE 72

4.5.29.1 Leak test or power supply battery leads: Add the following:

"Leak test of power supply battery leads. One battery with welded leads at the start of each shift and one of every 100 manufactured shall be tested for compliance with the gross and fine leak requirements of MIL-B-48761. Any unit failing to meet the requirement shall be classified defective and all units fabricated since the last production test are classified defective and shall be leak tested 100%. (Non-Destructive Test) (Alternate Lead Material Only.)".

5. PACKAGING: Delete "5.1, 5.2, 5.3 in their entirety" and substitute the following:

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"5.1 Packaging. The Electronics-MCD assembly shall be preserved and packaged in accordance with level B of MIL-P-14232. Each assembly shall be wrapped in aluminum foil, packaged in a fiberboard box and separated by loose partitions and separators as required. Cushioning as needed shall be used; the fiberboard box shall be sealed in a barrier bag conforming to Type I, Class F, Style 1, Spec. MIL-B-117. (Material used to fabricate the barrier bag must include metal foil). The packaging materials shall not contain silicones, sulphur compounds or polysulphides.

5.2 Packing. The Electronics-MCD assemblies in box and barrier bag shall be packed in accordance with Level B of MIL-P-14232 and Code of Federal Regulations, Title 49 with the following mandatory requirements: The outer packing box shall be wood or fiberboard conforming to the requirements of Code of Federal Regulation, Title 49. Sufficient cushioning and separators must be used between the barrier bag and outer box to prevent damage to the items or tearing of the barrier bag.

5.3 Marking. The marking of the outer box shall be in accordance with MIL-STD-129, and DOT marking per Code of Federal Regulation, Title 49. Marking indicating the components are fragile electronics shall be employed."

PAGE 73

FIGURE 1: Delete in its entirety.

PAGE 74

FIGURE 2a, b: Delete in its entirety.

PAGE 75

6.2 subpara a: Delete title of MIL-E-63399 and substitute "Electronics-MCD Assembly".

In title of MIL-P-63421 change "Fascom" to "FASCAM".

6.3:

Line 3: Delete "US ... Command" and substitute "U.S. Army Armament Research Development and Engineering Center".

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Line 4: Delete "DRDAR-QAT-I" and substitute "AMSMC-QAT-I(D)".

6.4:

Lines 2 and 3: Delete "US .. (ARRADCOM)" and substitute "US Army Armament Research Development and Engineering Center (ARDEC)".

Line 7: Delete "ARRADCOM" and substitute "ARDEC".

PAGE 76

6.5, lines 4 and 5: Delete "US ... DRDAR-QAT-M" and substitute "U.S. Army Armament Research Development and Engineering Center, ATTN: AMSMC-QAT-M (D)".

PAGE 78 (APPENDIX)

30.1, line 14: Delete "less than .015 inches .." and substitute "less than .010 inches ..".

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